

CO1 - Assessment - 03.

Name: C. Sangita

Reg NO: 192373059

Course code - ITAI435

Course : Ethical Hacking

A Nation-state-sponsored group breaches a hospital's systems to disrupt operations during a geopolitical conflict. Analyze this scenario across the categories of hacker types and explain why classification matters for response and law.

A Nation-state-sponsored Attack on a Hospital:

Introduction:

In the given scenario, a nation-state-sponsored group breaches a hospital's computer systems to disrupt its operations during a geopolitical conflict. Since the attackers are supported by a government and intentionally perform unauthorized and harmful activities, they can mainly be classified as state-sponsored hackers and black-hat hackers. The attack is more serious because hospital is a critical service where disruption can affect patient care and safety.

Hacker classification and scenario Analysis:

Hacker type	Main Purpose	Relation to the given case
White Hat	Authorized security testing	Not Applicable
Black Hat	Malicious and unauthorized activity	Applicable

Gray Hat	unauthorized access without necessarily causing serious harm	Not suitable
Hacktivist	Political / social objectives	Partially related
Cybercriminal	Financial gain	Not main purpose
State-sponsored Hacker	Gov - supported strategic / geopolitical attack	Main classification

→ The state sponsored category is the most appropriate because attackers are connected to nation-state & the attack occurs during a geopolitical conflict.

Attacks and its Effects:

Attack flow:

Geopolitical conflict



state sponsored Group



unauthorized access



Hospital network compromise



system disruption



healthcare services affected.

Major Effects:

- Patient records may be stolen or exposed.
- Medical data may be modified or deleted.
- Hospital computer systems may become unavailable.
- Emergency and appointment services may be disrupted.
- Doctors may lose access to important patient information.
- Patient treatment may be delayed.
- Confidentiality, integrity and availability (CIA) may be affected.
- Patient safety may be put at risk.

Why classification matters:

- correct classification helps determine the motivation, threat level, response, and legal action.

For Response:

- * Identifiers the motivation of the attacker.
- * Helps determine the severity of the threat.
- * Helps select the correct incident-response strategy.
- * State-sponsored groups may have advanced tools and greater resources.
- * Affected systems should be isolated immediately.
- * Digital evidence and system logs should be preserved.
- * The incident should be reported to appropriate authorities.
- * National cybersecurity agencies may need to be involved.
- * Cybersecurity teams should investigate the attack.

Why classification matters for Law:

- * unauthorized access can result in legal action
- * theft or exposure of patient data can have legal consequences.
- * Destruction or modification of hospital data can be an offence.
- * Disruption of essential healthcare services can lead to serious penalties.
- * state sponsorship can make the incident a national-security issue.
- * If a foreign government is involved, international cooperation may be required.
- * correct classification helps identify the appropriate legal authorities.

Justification:

- the attackers are state-sponsored hackers because they are supported by a nation-state.
- They are Black Hat hackers because their access is unauthorized and malicious.

Conclusion:

Finally classification is state-sponsored + Black Hat hacker. They attack threatens hospital systems, patient information, and patient safety. Therefore identifying the hacker type is essential for effective cybersecurity & law.